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Feeder use predicts both acquisition and transmission of a contagious pathogen in a North American songbird

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Individual heterogeneity can influence the dynamics of infectious diseases in wildlife and humans alike. Thus, recent work has sought to identify behavioural characteristics that contribute disproportionately to individual variation in pathogen acquisition (super-receiving) or transmission (super-spreading). However, it remains unknown whether the same behaviours enhance both acquisition and transmission, a scenario likely to result in explosive epidemics. Here, we examined this possibility in an ecologically relevant host–pathogen system: house finches and their bacterial pathogen, *Mycoplasma gallisepticum*, which causes severe conjunctivitis. We examined behaviours likely to influence disease acquisition (feeder use, aggression, social network affiliations) in an observational field study, finding that the time an individual spends on bird feeders best predicted the risk of conjunctivitis. To test whether this behaviour also influences the likelihood of transmitting *M. gallisepticum*, we experimentally inoculated individuals based on feeding behaviour and tracked epidemics within captive flocks. As predicted, transmission was fastest when birds that spent the most time on feeders initiated the epidemic. Our results suggest that the same behaviour underlies both pathogen acquisition and transmission in this system and potentially others. Identifying individuals that exhibit such behaviours is critical for disease management.

1. Introduction

Heterogeneity in exposure and response to pathogens can strongly influence the duration and intensity of disease epidemics [1]. Although traditional models of disease dynamics frequently treat individuals as homogeneous or randomly variable [2], in reality, a small percentage (20%) of a population is often responsible for the majority (80%) of pathogen transmission [3]. Such disproportionate contributions to acquiring or spreading pathogens have been termed ‘super-receiving’ [4] and ‘super-spreading’ [5], respectively. Given the importance of individual heterogeneity to infectious disease dynamics [1], there is growing interest in identifying which individuals will be super-receivers or super-spreaders in both human and wildlife populations [6–9].

Host behaviour is a strong driver of heterogeneity in pathogen acquisition or spread for numerous systems [10–14]. Among wildlife, variation in territoriality [10], social status [7], foraging [15], social network connectivity [14,16], grooming [17] and aggression [4,17] has been shown to underlie individual variation in exposure to infectious agents (super-receiving). Among humans, variation in behavioural traits such as hand hygiene [18] and social contacts [5] among healthcare workers can increase super-spreading in severe acute respiratory syndrome.

Similarly, high connectedness in human sexual networks can increase the chance of acquiring and spreading sexually transmitted disease [19]. However, behavioural predictors of super-spreading in wildlife remain largely unknown because of the lack of detailed contact tracing.

Although behaviour can be important for predicting individual variation in pathogen acquisition and spread in some systems, it remains unknown whether the same behaviours make an individual both more likely to be a super-receiver and more likely to be a super-spreader. In the case of Tasmanian devil facial tumour disease, biting conspecifics is a strong predictor for acquiring disease, while being bitten is not [4]. Thus, while aggressive Tasmanian devils are likely super-receivers, they are unlikely to be super-spreaders. In other systems, however, the same behaviours can influence both pathogen acquisition and transmission. For example, host defensive behaviours against arthropods can influence both exposure to vector-borne pathogens and the capacity to spread these pathogens [20,21]. In such systems, where the same individuals are both super-receivers and super-spreaders, behavioural covariation in exposure and infectiousness could have significant epidemiological consequences [22]. Here, we examine this issue through field and experimental studies in an emerging wildlife host–pathogen system: house finches (*Haemorrhous mexicanus*) and their bacterial pathogen *Mycoplasma gallisepticum* (MG).

Since the mid-1990s, the house finch, a common North American songbird, has been host to an emerging clade of MG [23], historically a poultry pathogen. In house finches, MG causes severe conjunctivitis [24] and has been associated with significant population declines [25]. Following MG's emergence in finches, annual winter epidemics have been observed throughout the USA [26]. This pattern is strongly associated with the social behaviour of non-breeding house finches: forming loose flocks and congregating around bird feeders [26,27].

The house finch–MG system provides a powerful context for testing how variation in behaviour predicts the risk of acquiring and spreading infection. House finches are highly social during the relevant period of MG transmission. Direct interactions between individuals are likely critical for transmission, because MG, like all Mollicute bacteria, lacks a cell wall, precluding it from surviving for long outside of its host [28]. Furthermore, indirect transmission via fomites (i.e. objects capable of harbouring infectious organisms) is also possible in this system [29]. Moreover, controlled experiments can be readily conducted [30], enabling us to quantify how different individual-level behaviours relate to pathogen transmission.

Here, we examine three classes of behaviours potentially associated with super-receiving and super-spreading in this system: social network position, aggressive interactions at feeders and foraging behaviours. First, given that gregarious individuals likely come into contact with more conspecifics, we predicted that house finches most central in the social network and those associating in larger flocks would more likely act as super-receivers, via direct or indirect transmission. Second, because agonistic interactions are the primary source of direct contacts during the non-breeding season (when MG prevalence peaks) [27], we predicted that birds with more frequent or diverse displacements at feeders would be more likely to acquire disease if direct transmission is important to transmission. Third, given that feeders can serve as fomites of MG [29], we predicted that finches using feeders more often would be more likely to show conjunctivitis if indirect transmission is most important in this system. We tested these

predictions in free-living house finches using radiofrequency identification (RFID) and repeated captures over the course of five months. After determining the behavioural traits associated with super-receiving in the field, we initiated replicated experimental epidemics in captive flocks by inoculating 'index' birds exhibiting different behaviour to assess whether the behaviours associated with super-receiving are also associated with super-spreading.

2. Methods

(a) Field study

(i) Initial captures and temporary housing

From October 2012 to February 2013, we captured house finches across six sites on and near Virginia Tech's campus using baited traps and mist nets. Trapping occurred twice per week (only once per week at each site). Sites were close enough that birds could use all six (maximum distance between sites = 2.3 km), though most individuals used fewer (median = 2, range = 1–5). We captured and marked 180 individuals, of which 117 were detected via RFID at least once and 35 were physically recaptured (median times recaptured = 1, range = 1–3). We estimated the local population size to be 364 (see the electronic supplemental material), suggesting that we marked roughly 49% of the population, with 32% of the population ($n = 117$ detected via RFID) contributing to our social network (see below). Recent work suggests that marking 30% of a population of this size likely yields accurate metrics of an individual's relative connectedness within a social network [31].

Birds were housed overnight for a separate study (see the electronic supplemental material), fitted with plastic leg-bands, a US Fish and Wildlife Service aluminium band and a unique passive integrated transponder (PIT) tag, which was secured to plastic bands using electrical tape [32]. Similar tags do not affect fitness in other small passerines [33]. No detached PIT tags were detected during the study. Birds were released at their capture site within 22–30 h.

(ii) Assessing infection

All birds were scored for clinical signs of mycoplasmal conjunctivitis on an 'eye score' scale (0–3 per eye) as described elsewhere and in the electronic supplementary material [30]. Eye score provides a non-invasive, accurate proxy for infection: we tested a subset of wild-caught animals for the presence of MG using qPCR and found strong agreement with eye score (electronic supplementary material, figure S1).

(iii) Monitoring passive integrated transponder-tagged birds

At all sites, we installed a bird feeder equipped with an RFID system [34] to log visits by PIT-tagged individuals. Feeders were tube-type, with two feeding ports. Below each port, we placed an RFID antenna, connected to a reader that logged one data-point per second. Batteries were changed regularly to ensure uninterrupted logging.

(iv) Extracting behavioural metrics from radiofrequency identification data

We quantified three classes of behaviours (social networks, aggressive interactions and foraging behaviours) using our RFID data. Although house finches also have social interactions at non-feeder locations, house finches in the eastern US depend heavily upon feeders [35–37]. These sites are the primary source of contact during the non-breeding season, suggesting that behavioural interactions at feeders are particularly relevant to pathogen

transmission. Moreover, numerous studies have shown that flock membership inferred from feeders predicts the spread of information through populations and captures broader social interactions relevant to fitness [38,39]. Although single instances of co-occurrence contain little information for constructing social networks, analysing aggregate patterns of co-occurrence across many instances (in our case, thousands of records) is key to generating robust networks [40,41].

Social networks. To assess whether social networks predicted the risk of conjunctivitis, we first applied a machine learning algorithm (Gaussian mixture model, [42]) to identify co-feeding events. This algorithm identifies clusters of detections on feeders, by evaluating the non-uniform (or 'bursty') data-stream, circumventing the need to use arbitrary time thresholds when defining associations. We next used the R package *asnipe* [43] to generate a network based on patterns of co-occurrence by individuals in the same feeding events. We then defined associations between birds (network edges) using the simple ratio index. This represents the probability of observing two individuals together given that at least one was observed (e.g. 0 for dyads never observed together, 1 for dyads always observed together).

Using these foraging networks, we estimated each bird's position within the social network using weighted degree and eigenvector centrality in the *sna* package for R [44]. These metrics reflect a bird's local connectedness and network-wide connectedness, respectively, and are predictive of information acquisition [38] and infection in other wildlife systems [6–8]. Because flock size has been associated with higher prevalence of mycoplasmal conjunctivitis in finches [26], we calculated each bird's average adjusted group size. Adjusted group size was defined as the number of marked individuals in each foraging flock divided by the total number of marked individuals detected that day.

Aggressive interactions. To quantify the propensity for direct contact among conspecifics, we quantified displacement events at feeders. We defined displacement as the detection of two birds at the same feeder port within 2 s of one another. For each individual, we calculated the average number of total displacements in which that bird was involved per day, as displacing or being displaced could equally lead to physical contact. We also calculated the average number of unique individuals per day with which a focal bird was involved in displacements. Finally, we used displacements to calculate relative dominance, using the residuals of a regression between times displaced versus times displacing. A positive relative dominance value means that an individual displaced others more often than it was displaced.

Foraging behaviours. We extracted three metrics of foraging behaviour hypothesized to be relevant for acquiring MG: (i) the total amount of time individuals were logged at feeders per day, (ii) the number of unique feeders visited per day and (iii) the total number of feeding bouts individuals made, assuming a 4 s or longer gap in detection of the same individual indicated separate bouts.

(v) Statistical analyses: field study

Because we relied on repeated observations of individuals to robustly quantify behaviours relevant to disease acquisition, and we had a significant proportion of transient individuals in our population, we limited our final analysis of disease risk to individuals that were detected by RFID on 7 or more days ($n = 76$, 35 of whom were also physically captured multiple times) [45]. However, because transient birds still formed part of the flocks, we included all individuals ever detected by RFID ($n = 117$) when constructing the social network (including identifying groups) and generating metrics on aggressive interactions.

To determine which behaviours are associated with mycoplasmal conjunctivitis in the wild, we used generalized linear models with binomial error distributions in R [46]. The dependent variable was whether an individual was ever detected as diseased (1) or not (0). Models were compared and parameters

averaged using Akaike information criterion corrected for small sample sizes (AICc) [47] in the MuMIn package for R [48]. Relative importance of each parameter was calculated by summing the AICc weights from all models in which that variable occurred. Independent variables are listed in table 1. For each bird, rates were calculated as a total from the entire study divided by the number of days on which that bird was detected. The number of times captured was included in every model to control for potential detection bias whereby birds captured more often could be more likely to be seen with conjunctivitis. Finally, because multicollinearity in a model averaging framework can lead to biased parameter estimates with correlations as low as 0.55–0.65 [49], several pairs of variables (correlations >0.55 , see electronic supplementary material, table S1) were never included in the same models. However, results of model averaging were qualitatively similar if this cut-off was increased to a more liberal 0.7.

Because null hypotheses when using social network metrics are not necessarily 'no effect' [41,50], we used randomization tests to determine if parameter estimates for social network variables in top models ($\Delta\text{AICc} < 2$) differed significantly from random. We used the *asnipe* package for R [43], where for each permutation step (1–10 000) the observations of individuals in two different groups were swapped, creating an increasingly random network [51,52]. We fit GLMs using the weighted degree or eigenvector centrality calculated for each random network with the same fixed effects for each of the top models, creating a distribution of coefficient values for these metrics [53]. If the coefficient in the observed network exceeded 95% of the coefficient values from the randomized networks, we considered this as support for a role of the tested metric in influencing disease risk.

(b) Experimental epidemics in captivity

(i) Experimental design

Because our field data indicated that time spent on feeders was the most important predictor of super-receiving in this system (see Results), we designed experimental epidemics in captivity to test whether time spent on feeders also predicted super-spreading. We quantified transmission dynamics in 10 single-sex flocks that differed only in the behaviour of the initially infected individual (electronic supplementary material, table S2). In half of the flocks, we initiated an epidemic by infecting the individual that spent the most time on the feeder (*high-feeding index*); in the other half, we infected the individual that spent the least time on the feeder (*low-feeding index*).

(ii) Field captures, transport and pathogen history assessment

During March 2014, we captured house finches at two sites in and near Tempe, AZ. No birds showed clinical signs of mycoplasmal conjunctivitis, which has not been reported in this region. Two weeks after arriving at Virginia Tech, all birds used in this experiment ($n = 44$) were negative for anti-MG antibodies [54] (electronic supplementary material).

(iii) Flock housing

In May 2014, birds were fitted with plastic leg bands and PIT tags, and transferred to single-sex flocks of four, in 76 cm $\times$ 84 cm $\times$ 46 cm cages harbouring one tube-type feeder with one feeding port. Flocks of four regularly occur in the wild, particularly in the southeastern USA (mean winter flock size: 4–8), where MG epidemics have been explosive [26]. Feeder perches were equipped with an antenna and RFID receiver. Plastic sheets were draped between cages to minimize cross-cage transmission. Baseline data on feeder use were collected for 7 days.

(iv) *Mycoplasma gallisepticum* inoculation and treatment groups

After two weeks in flocks, one bird per flock ('index') was inoculated in each conjunctiva with 40 μl of a solution of Frey's

Table 1. Model-averaged parameter estimates and relative importance of variables in generalized linear models of the risk of mycoplasmal conjunctivitis in wild house finches. Normalized model-averaged parameter estimates are displayed for visual comparison.

model parameter	model-averaged parameter estimate (95% CI)	relative importance	normalized model-averaged parameter estimate (95% CI)
intercept	−4.29 (−8.04 to 0.54)	n.a.; contained in all models	
number of captures	0.82 (−0.25 to 1.89)	n.a.; contained in all models	
time spent on feeders per day	0.17 (0.019–0.32)	0.87	
weighted degree	−3.30 (−7.46 to 0.87)	0.53	
average adjusted group size	21.90 (−28.54 to 72.34)	0.34	
eigenvector centrality	−3.30 (−22.62 to 16.04)	0.33	
relative dominance	0.55 (−4.33 to 5.43)	0.25	
number of unique individuals w/which interacted aggressively per day	−0.61 (−3.78 to 2.55)	0.24	
sex (males)	−1.09 (−2.89 to 0.71)	0.18	
(unknown, $n = 1$)	−14.13 (−4800 to 4772)	—	
feeders visited per day	−0.61 (−10.60 to 9.39)	0.09	
feeding bouts per day	0.10 (−0.020 to 0.23)	0.05	
instances of aggression per day	0.44 (−0.14 to 1.02)	0.01	

medium containing 2×10^5 colour changing units of MG, diluted from a stock described in the electronic supplementary material.

Index birds were chosen based on the time they spent at the feeder during the prior week. We inoculated either the bird that spent the most time on the feeder (*high-feeding index* flocks) or the bird that spent the least time on the feeder (*low-feeding index* flocks). Because within-group variation in time spent on the feeder differed among flocks, we balanced our treatments across low-, mid- and high-variance flocks (electronic supplementary material, table S2). Additionally, some flocks had consistent high-feeders, whereas others had consistent low-feeders. Because these flocks appeared at equal rates across low-, mid- and high-variance groups, we inoculated low-feeders in flocks with consistent low-feeders and high-feeders in flocks with consistent high-feeders (electronic supplementary material, table S2).

One flock of three birds and one single-housed bird were left uninfected to serve as sentinels for cross-cage transmission (electronic supplementary material, table S2). Owing to the ethical and logistic limitations of keeping wild birds in captivity, the number of sentinels was low, but none showed detectable pathogen load or eye score.

(v) Monitoring the experimental epidemic

One day prior to inoculation of index birds, and every 2 days post-inoculation (PI) until day 22 PI, all animals were captured for sampling. Sampling included eye scoring for clinical signs of infection (0–3 point scale described in the electronic supplementary material), and conjunctival swabbing for quantification of MG load by qPCR, using previously published methods ([54]; see the electronic supplementary material).

We used two methods to determine the initial transmission event within each flock. First, based on eye scores, initial transmission was defined as the first day on which any non-index bird showed eye score above 0. Second, because it is unknown whether transmission of MG can occur when birds are subclinical and because low false-positive values can occur with our highly sensitive qPCR assay, we used two different qPCR-based cut-offs. These analyses defined initial transmission as the first day on which any non-index bird showed pathogen load above a given number of copies of the *mgc2* gene. The first cut-off value was one copy; the second was 1349 copies, which reflects an estimate of the minimum infectious load in this system (see the electronic supplementary material).

(vi) Statistical analyses: experimental epidemics

To determine whether birds that spend the most time on feeders act as super-spreaders, we tested whether transmission was more rapid in our *high feeding index* flocks using accelerated time failure models with a Weibull distribution in the ‘survival’ package [55] for R. This distribution makes minimal assumptions about the risk of transmission over time [56]. In addition, this distribution better fit the data than did other distributions tested (exponential, Gaussian, logistic, lognormal, loglogistic or t), as assessed by $\Delta AICc > 2$ [47]. To control for possible sex effects on transmission, we included sex as a covariate. In this analysis, the unit of replication was the flock ($n = 10$).

To assess whether high- and low-feeding index birds were consistent in their feeding behaviours across time, we calculated repeatability using the intraclass correlation coefficient (ICC) in package ICC in R [57]. Finally, to test whether high- and low-feeding index birds might differ in other transmission-relevant metrics, we used general linear-mixed effects models in package nlme in R [58] to test whether the two types of index birds displayed different levels of pathogen load or eye score across time.

3. Results

(a) Field observations

(i) Social network position

We found limited support, in the opposing direction of our prediction, that social network position predicts risk of mycoplasmal conjunctivitis. In particular, higher weighted degree was associated with a reduced risk of conjunctivitis. Weighted degree had a relative importance of 0.53 (table 1; see electronic supplementary material, table S3 for the top 10 models), but the 95% confidence interval for this variable overlapped zero, suggesting a weak overall effect. However, among the top three models ($\Delta AICc < 2$; electronic supplementary material, table S3), two included weighted degree and permutation-based p -values for weighted degree in both models were less than 0.01. This suggests that although the size of the effect may be small, higher weighted degree may be associated with reduced risk of conjunctivitis. We detected potential, but weak, effects of average adjusted group size and eigenvector centrality (relative importance of 0.34 and 0.33, respectively; both parameter estimates with 95% confidence intervals overlapping zero). Eigenvector centrality did not occur in any of the top three models ($\Delta AICc < 2$), but did appear in two of the top 10 models (electronic supplementary material, table S3). Although permutation-based p -values for eigenvector centrality from these models were both less than 0.01, the signs of the parameter estimates were opposite, suggesting weak or inconsistent effects. Overall, results from social network metrics suggest that birds in larger groups may have had higher risk of mycoplasmal conjunctivitis, whereas birds more central in the network may have had lower risk.

(ii) Aggressive interactions

We found minimal support for our prediction that birds engaged in frequent or diverse aggressive interactions are at higher risk for acquiring mycoplasmal conjunctivitis. Each metric of aggressive contacts had a relative importance ≤ 0.25 and 95% confidence interval overlapping zero (table 1).

(iii) Foraging behaviours

We found strong support for our prediction that foraging behaviours are important for disease risk in this system. The total time that free-living house finches spent on feeders per day positively predicted the probability of mycoplasmal conjunctivitis, with a relative importance of 0.87, higher than any other variable (figure 1*a* and table 1). Moreover, total time spent on feeders was the only variable whose 95% confidence interval did not overlap zero. Exclusion of the bird that spent the most time on feeders yielded nearly identical results (figure 1*b*).

(iv) Experimental epidemics

When transmission was assessed using eye score, *high-feeding index* flocks, those in which the bird that spent the most time on the feeder was inoculated, showed a significantly shorter time to initial transmission than did *low-feeding index* flocks, in which the bird that spent the least time on the feeder was inoculated (figure 2*a*; overall model: $\chi^2_{d.f.=2} = 11.69$, $p = 0.003$; treatment (low): parameter estimate = 0.48, $z = 5.42$, $p < 0.001$). This result held true when controlling for the

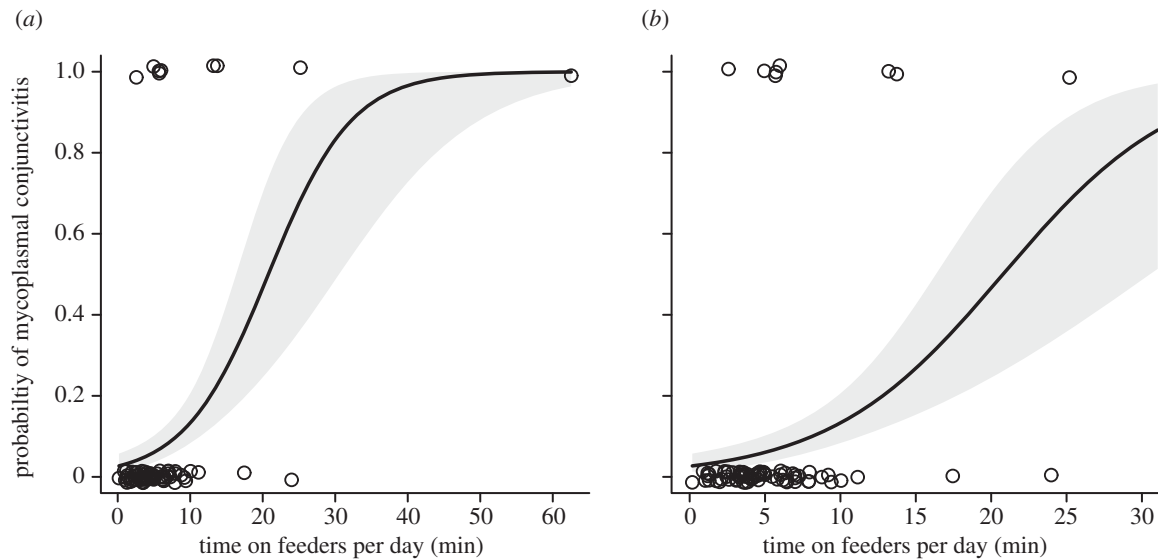


Figure 1. Free-living house finches that spend more time on bird feeders are more likely to show mycoplasmal conjunctivitis (a). Removal of the rightmost point yields nearly identical model predictions (b). Lines show predictions based upon top models from AICc model selection ± 1 s.e. (shaded area).

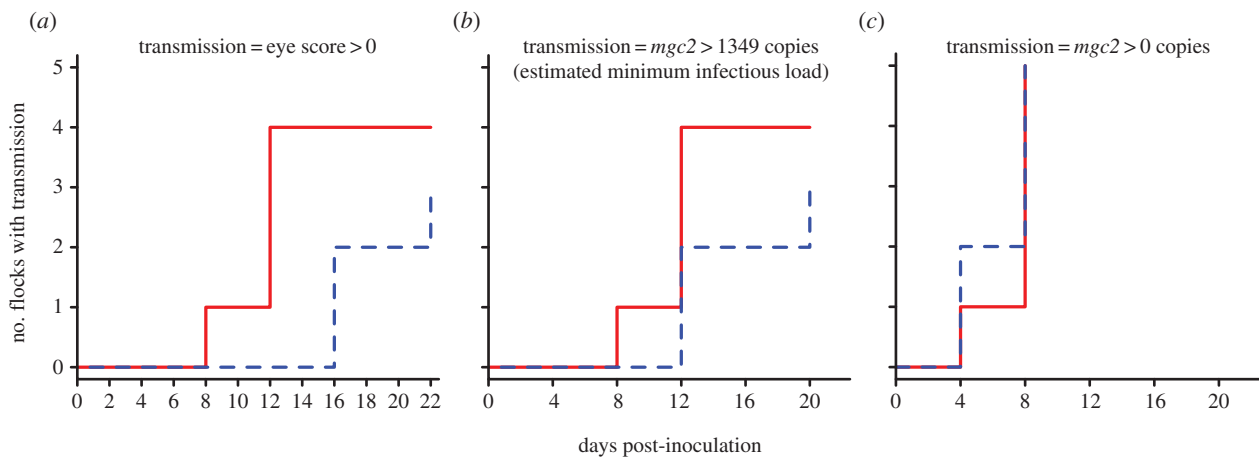


Figure 2. In captive flocks of house finches, epidemics of *M. gallisepticum* progress more rapidly in flocks where the bird spending the most time on the feeder is experimentally inoculated ('high-feeding index' flocks, solid red line) than in flocks where the bird spending the least time on the feeder is inoculated ('low-feeding index' flocks, dashed blue line). When pathogen transmission is defined as the first occurrence of clinical signs (a) or the first detection of the estimated minimum infectious pathogen load (b) in an originally naive animal, this pattern is highly significant. However, when transmission is defined as any detectable pathogen load, no differences between groups appear (c). (Online version in colour.)

effect of sex, as male flocks tended to have more rapid transmission than did female flocks (sex (male): parameter estimate = -0.64 , $z = -3.06$, $p = 0.002$).

Differences in transmission between groups as measured by pathogen load were sensitive to the cut-off used. When using our best estimate of minimal infectious pathogen load (1349 copies of the *mgc2* gene; see the electronic supplementary material), results closely match those obtained when defining transmission using eye score (figure 2b, overall model: $\chi^2_{d.f.=2} = 9.39$, $p = 0.009$; treatment (low): parameter estimate = 0.35 , $z = 2.70$, $p = 0.007$; sex (male): parameter estimate = -0.61 , $z = -2.58$, $p = 0.01$). However, using a cut-off of one copy of the *mgc2* gene, no differences in transmission between groups were evident (figure 2c, overall model: $\chi^2_{d.f.=2} = 0.68$, $p = 0.71$; treatment (low): parameter estimate = -0.05 , $z = -0.50$, $p = 0.62$; sex (male): parameter estimate = 0.09 , $z = 0.79$, $p = 0.43$).

Individual birds were consistent in their relative levels of feeder use both before and after inoculation (ICC; (experimentally inoculated birds only) = 0.67 ; electronic supplementary material, figure S2; ICC (including all birds) = 0.67 , data not

shown). In addition, high- and low-feeding index birds did not differ in the magnitude or time course of eye score or pathogen load (electronic supplementary material, figure S3; eye score: time: $F_{12,96} = 15.65$, $p < 0.001$, treatment: $F_{1,8} = 0.07$, $p = 0.80$, time $\times$ treatment: $F_{12,96} = 0.36$, $p = 0.97$; pathogen load: time: $F_{3,24} = 79.36$, $p < 0.001$, treatment: $F_{1,8} = 0.57$, $p = 0.47$, time $\times$ treatment: $F_{3,24} = 0.67$, $p = 0.58$).

4. Discussion

Through observation in the field and captive experimentation, we found that individual variation in feeder use predicted both the likelihood of house finches acquiring mycoplasmal conjunctivitis in the wild and the likelihood of transmitting the causative agent of this disease, MG, in captivity. These results provide an example of a wildlife system in which the same behaviour makes an individual likely to be both a super-receiver and a super-spreader.

Such consistency in the traits underlying both exposure and transmission should favour more rapid and severe epidemics.

Consistent with prior work in this system [29,59], our field results suggest that feeders play an important role in the risk of acquiring mycoplasmal conjunctivitis. Hartup *et al.* [59] found that the presence of tube-style bird feeders was associated with an increased prevalence of mycoplasmal conjunctivitis at the backyard scale. Our data illustrate a similar pattern at the individual level: the extent to which a house finch interacts with feeders in the wild predicts its likelihood of displaying mycoplasmal conjunctivitis. These results are also consistent with Dhondt *et al.* [29] finding that feeders can act as fomites, facilitating indirect transmission of MG.

Although interaction with feeders was positively related to the risk of mycoplasmal conjunctivitis in the wild, we found little evidence to suggest that aggressive interactions (a proxy for direct contacts), or location within a social network, which could reflect direct or indirect contacts, were strongly predictive of disease risk. Consistent with our predictions and prior work [26], birds in larger groups tended toward higher risk for mycoplasmal conjunctivitis, though the model-averaged parameter overlapped zero, suggesting a weak effect. Non-intuitively, we found that birds with lower weighted degree were more likely to be seen with mycoplasmal conjunctivitis. Taken together, these metrics suggest that birds foraging in larger groups with unstable membership (low weighted degree, despite large sizes) are the most likely to acquire mycoplasmal conjunctivitis, but the relative contributions of direct versus fomite-based contacts to this pattern remain unknown. An alternative explanation for why birds with lower weighted degree were more often seen with conjunctivitis is that healthy individuals avoided diseased individuals. In this scenario, avoidance of sick birds would drive down diseased individuals' weighted degrees and eigenvector centralities. While one study of captive house finches showed that healthy individuals actually prefer feeding near MG-infected conspecifics [60], another study found that finches can avoid birds actively mounting an inflammatory immune response [61]. Thus, avoidance of infected individuals cannot be ruled out.

Overall, our results suggest that time spent on bird feeders plays a more critical role in MG dynamics in wild birds than social network position or aggressive interactions. Constraints on our sampling regime might have limited our power to detect subtle effects of social network position on MG risk. Moreover, if behaviour away from feeders (e.g. at roosting sites) is unrelated to flock membership during foraging, our sampling may miss important social connections. However, prior work found that radio-tagged house finches roosting together at night were more likely to be seen together during the day [62], suggesting that our social metrics should also reflect behaviours away from feeders.

Our field data lack continuous information on the disease status of all individuals, limiting our ability to assess causal relationships between feeding behaviour and disease risk. Although we physically captured approximately half of the birds in that study at least twice ($n = 35/76$), only three individuals changed disease status between captures. Thus, we lacked the resolution necessary to eliminate the possibility that post-infection behavioural changes altered the relationship between feeding and conjunctivitis. This remains an open research question in the study of disease dynamics in the wild. However, data from our captive experiment support time on the feeder as a cause (rather than a consequence) of

increased risk of conjunctivitis: the feeding behaviour of inoculated individuals remained consistent prior to and following inoculation (electronic supplementary material, figure S2). This pattern suggests that post-infection behaviour is unlikely to disproportionately influence the correlation between feeder use and disease risk in the wild, though more intensive study is needed on this topic.

While our field study shows that feeding behaviour predicts the likelihood of acquiring disease in the wild, our captive experiment demonstrates that feeding behaviour of infected individuals also predicts the likelihood of transmitting MG to susceptible flockmates. When transmission was assessed via visible pathology or the estimated minimum transmissible pathogen load, MG spread significantly more rapidly within flocks in which the inoculated bird spent the most time on the feeder. While differences in pathogen load or pathology, both of which alter pathogen deposition on feeders [63], could have influenced the pattern of transmission, neither factor differed between high- and low-feeding index birds. Additionally, although dominance status correlated with time spent on feeders, it is unlikely that dominance *per se* resulted in more rapid transmission. In support of this idea, when models of time to transmission containing dominance were compared with models containing time on the feeder, the latter always showed lower AICc values (electronic supplementary material, table S4). Taken together, these results indicate that feeding behaviour is the most parsimonious driver of the observed differences in transmission rates between treatments. Although behaviour has been hypothesized to impact disease dynamics in a suite of systems [10–14], this study provides rare empirical evidence that the behaviour of initially infected hosts can alter the trajectory of experimental epidemics in an ecologically relevant disease.

When we expanded our definition of transmission in captivity to include birds with one or more copies of the pathogen, we saw no differences in transmission between treatments. However, two factors suggest that such measurements are overly conservative. First, infectiousness (the ability to infect others) rather than merely harbouring a pathogen is the most relevant currency for transmission dynamics. Thus, assessing transmission using the estimated minimum infectious load should better capture the dynamics of our experimental epidemics. Second, two individuals were detected with loads of less than 50 copies of MG at only one time point. These transient, low pathogen loads could reflect false positives in our qPCR assay or low pathogen burdens that were likely cleared before the bird could become infectious. In either case, counting these events as successful transmission likely overstates their importance in the epidemics.

Taken together, our field and captive results suggest that in this system, the same behaviour (feeder use) is important for both acquisition and transmission. These results highlight the importance of understanding relationships among behaviours that contribute to pathogen acquisition and spread. Specifically, the pathogen's potential for transmission should be highest, and epidemics most explosive, in systems with positive correlation between behaviours that predict acquisition and transmission [64]. In such cases, the highest risks of both acquiring and spreading a pathogen would be concentrated in the same individuals. For such systems, targeted disease management could be highly efficient and effective [10], as only one subset of individuals would need to be identified. In contrast, without correlation between

the risks of acquisition and spread, similar management would require identifying two unique subsets of high-risk individuals, or focusing efforts on one class or the other, potentially limiting the efficacy of intervention. Further empirical and theoretical efforts are needed to describe such behavioural correlations and their impacts on disease dynamics across systems.

Ethics. Studies were conducted under these permits: Virginia Tech IACUC (10-059-BIOL and 13-090-BIOL), Virginia Department of Game and Inland Fisheries (038781 and 044569), Arizona Game and Fish Department (SP654946 CLS), United States Fish and Wildlife Service (MB158404-1), and United States Geological Survey Bird Banding Laboratory (23513). Animals used in captive experiments were euthanized using an overdose of the anaesthetic, isoflurane, as approved by the Virginia Tech IACUC.

Data accessibility. Field and captive experiment data including disease metrics and radiofrequency identification detected behavioural data

are deposited in the Dryad repository: <http://dx.doi.org/10.5061/dryad.4ht07>.

Authors' contributions. J.S.A., S.C.M. and D.M.H., designed and executed observations and experiments and wrote the paper. J.S.A. and D.R.F. performed the analyses.

Competing interests. We declare we have no competing interests.

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